

Libraries, Modules & Text as Data

Pre-Assessment 5 · 11 questions · 39 points

Name: _____

Date: _____

Score: / 39

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write **“not yet”** and move on.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Three ways to import

3 points

Write the line of code that uses each import to do the job.

Import	Job	Your line of code
<code>import math</code>	square root of 16	
<code>from random import choice</code>	pick from <code>["a", "b"]</code>	
<code>import matplotlib.pyplot as plt</code>	bar chart of <code>x</code> and <code>y</code>	

2. kidtools.py

3 points

FILE: KIDTOOLS.PY

```
def shout(text):
    return text.upper() + "!"
```

FILE: MAIN.PY

```
import kidtools
from kidtools import shout
print(kidtools.shout("hi"))
print(shout("bye"))
```

(a) What does `main.py` print?

(2 pts)

(b) Where must `kidtools.py` be saved for `import kidtools` to work?

(1 pt)

3. Library words

4 points

Meanings:

1. A single `.py` file of functions and classes.
2. A folder of modules that belong together.
3. An online collection of free Python libraries anyone can install.
4. The tool that downloads and installs third-party libraries.
5. A big collection of code for one purpose, like matplotlib.

(a) Write the number of the matching meaning next to each word.

(3 pts)

Word	Meaning number
module	
package	
PyPI	

(b) Write the command you would type in the terminal to install the `rich` library.

(1 pt)

4. Read the docs

3 points

FROM THE PYTHON DOCS

```
random.randrange(start, stop, step=1)
    Return a randomly selected number from range(start, stop, step).
```

(a) What is the smallest number of arguments you must give?

(1 pt)

(b) What value does `step` have if you leave it out?

(1 pt)

(c) Write a call that picks a random **even** number from 0 to 10 (including 10).

(1 pt)

5. Turtle Art Studio

4 points

```
import turtle
t = turtle.Turtle()
for i in range(4):
    t.forward(100)
    t.left(90)
```

(a) What shape does this draw?

(1 pt)

(b) Which two numbers would you change, and to what, so it draws a triangle? (2 pts)

(c) `t` is an _____ of the `Turtle` class.

(1 pt)

6. Class Survey Dashboard

4 points

Fill in the four blanks to draw a labeled bar chart.

```
import matplotlib.pyplot as plt
pets = ["dogs", "cats", "fish"]
votes = [12, 9, 4]
plt._____(pets, votes)
plt._____("Favorite Pets")
plt.xlabel("Pet")
plt._____("Votes")
plt._____()
```

7. Loading & Cleaning Text

4 points

```
import string
text = "The cat sat. The cat ran!"
for p in string.punctuation:
    text = text.replace(p, "")
words = text.lower().split()
print(words)
print(len(words), len(set(words)))
```

(a) Write what the first print shows.

(2 pts)

(b) Write what the second print shows.

(1 pt)

(c) Lexical diversity = unique words ÷ total words. What is it for this text? (A fraction is fine.)

(1 pt)

8. Top Words

4 points

```
from collections import Counter
words = ["the", "fox", "and", "the", "dog", "fox",
        "the", "dog", "the", "fox", "cat"]
stop_words = {"the", "and", "a"}
counts = Counter(words)
print(counts["the"])
print(counts.most_common(1))
kept = [w for w in words if w not in stop_words]
print(Counter(kept).most_common(2))
```

(a) Write exactly what the program prints.

(3 pts)

(b) Why do word counters usually skip stop words?

(1 pt)

9. Author Fingerprint

3 points

```
book1 = {"dragon", "castle", "sword", "king"}
book2 = {"castle", "ship", "king", "storm"}
```

Write the words in each result (any order), or the number.

Expression	Result
book1 & book2	
book1 - book2	
len(book1 book2)	

10. Bug Safari

2 points

```
print("Mr. Smith is here. Hi!".split(". "))
```

(a) What does this print?

(1 pt)

(b) Why is this a bad way to split text into sentences?

(1 pt)

5 points

- opens the text file; if it is missing, prints `File not found` and returns an empty list
- makes the text lowercase and removes punctuation (use `string.punctuation`)
- splits it into words and **returns** the `n` most common words using `Counter`

```
[('the', 12), ('wolf', 7), ('pig', 6)]
```

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